

Multi-Horizon Tree-Neural Hybrid with Temporal Entity Propagation

1. Approach Overview

Detecting payment fraud under severe class imbalance (1.76% prevalence) and temporal regime drift requires moving beyond standard decision trees. Our solution combines three synergistic pillars:

1. **Causally Sound Behavioral Engineering:** 160 scale-free, strictly-prior features capturing individual spending baselines, short-term velocity acceleration, and bipartite entity novelty.
2. **Dual-Horizon Tree-Neural Hybrid:** Blending leaf-wise gradient-boosted trees (LightGBM) across two distinct training horizons with a continuous deep neural manifold (Tabular ResNet) to resolve orthogonal decision-boundary artifacts.
3. **Unsupervised Temporal Prediction Propagation:** A post-inference leave-one-out (LOO) diffusion process that exploits empirical transaction burst clustering without label leakage.

2. Data Handling & Causally Sound Feature Engineering

Strict temporal causality is enforced throughout: for every transaction at time t , engineered features access information strictly before t (broken by `transaction_id`). All features mechanically pass point-in-time leakage assertions; no per-entity target encoding or future information is used. From 220+ initial candidates, feature pruning isolated the **Top 160 core features**:

- **Scale-Free Behavioral Deviations:** Rather than raw amounts (which drift), amount is normalized against personal history:
 - `cust_amt_ratio`: Amount divided by the customer's prior expanding mean (single highest-gain feature).
 - `cust_amt_robust_z`: Robust MAD-based z-score around expanding median:
 $(amt - median)/(MAD + \epsilon)$.
- **Traffic-Matched Trailing Windows:** Sub-hour and trailing windows (5m, 15m, 30m, 1h, 3h, 6h, 12h, 24h, 72h, 168h) using `closed="left"`. High-throughput devices and merchants leverage sub-hour windows to catch automated credential-stuffing bursts.
- **Velocity Burst Acceleration:** Normalized velocity ratios like `cust_burst_accel` ($(cnt_{1h} \times 24)/(cnt_{24h} + 1)$) detect abrupt spikes relative to recent personal baselines.
- **Counterparty Expansion & Novelty:** Tracks real-time account takeover signals (`is_new_device_for_customer`, `is_new_location_for_customer`) and shared device concentration (`device_id_nunique_customer_for_device_prior`).

3. Model Architecture: Dual-Horizon Tree-Neural Hybrid

Empirical audits revealed an acute summer regime shift: modern attacks shifted heavily toward device-velocity bursts rather than customer-amount anomalies. To adapt without discarding rare historical categoricals, we deployed a hybrid ensemble:

- **Dual-Horizon LightGBM Trees:**
 - *Full-Train Horizon (Jan 1 – Jul 15)*: 882 boosting rounds, `num_leaves=127`, `lr=0.02`, `feature_fraction=0.85`, `min_data_in_leaf=50`. Captures macro patterns and long-tail categories.
 - *90-Day Specialist Horizon (Apr 16 – Jul 15)*: 433 boosting rounds, `num_leaves=63`, `lr=0.02`, `feature_fraction=0.75`, `min_data_in_leaf=100`. Discards stale winter relationships and focuses on recent summer velocity attacks.

- *Horizon Blend*: $P_{\text{LGBM}} = 0.48 \cdot P_{\text{Full}} + 0.52 \cdot P_{90\text{d}}$ (Local PR-AUC: **0.5284**).
- **Tabular ResNet (Neural Manifold Regularization)**:
 - *Architecture*: Continuous numerical features are standardized (StandardScaler, clipped to $[-5, 5]$); categorical features pass through learnable entity embeddings ($d \approx 1.6 \times \text{card}^{0.56}$). The combined representation feeds into 3 Residual Blocks (LayerNorm $\rightarrow$ Linear(256, 512) $\rightarrow$ GELU $\rightarrow$ Dropout(0.15) $\rightarrow$ Linear(512, 256) $\rightarrow$ Dropout(0.15) $\rightarrow$ Skip Connection).
 - *Training*: Trained for 8 epochs using AdamW (lr=1e-3, wd=1e-4) with CosineAnnealingLR.
 - *Tree-Neural Blend*: $P_{\text{raw}} = 0.88 \cdot P_{\text{LGBM}} + 0.12 \cdot P_{\text{ResNet}}$ (**0.5298** PR-AUC). The continuous neural manifold smooths orthogonal tree step boundaries, rescuing borderline false negatives.

4. Post-Processing: Temporal Entity Prediction Propagation

Fraud in payment rails is inherently bursty. Analysis revealed strong empirical sibling clustering: $P(\text{sibling fraud} \mid \text{current fraud})$ is **3.87x baseline** on devices and **2.63x** on customers. To reflect this, we apply an unsupervised leave-one-out (LOO) temporal diffusion process over a sliding ± 60 -minute window ($|t_j - t_i| \leq 30 \text{ min}, j \neq i$):

$$P_{\text{final}}(i) = (1 - w) \cdot P_{\text{raw}}(i) + w \cdot \frac{1}{2} [\bar{P}_{\text{LOO, cust}}(i) + \bar{P}_{\text{LOO, dev}}(i)]$$

with $w = 0.50$. Transactions without in-window siblings (singletons, 95.3% of test rows) pass through completely unchanged; the blend refines only the 4.7% of transactions occurring in bursts. The operation touches zero labels and fits zero test parameters. This elevates high-confidence burst clusters and suppresses isolated noise, driving local PR-AUC to **0.5359** and Kaggle Public LB to **0.56548**.

5. Validation Strategy, Technical Decisions & Results

- **Validation Discipline**: Evaluated on a chronological holdout tail (**July 1 – July 15**, 59,465 transactions). Seed noise was measured at $\sigma \approx 0.0020$; innovations were retained only if lifts exceeded this noise floor.
- **Key Negative Findings (Avoided Traps)**:
 - *Class Weighting*: Removing `scale_pos_weight=55` gave +0.026 LB. Weighted loss distorts ranking calibration under PR-AUC.
 - *Forward Features*: Discarded forward-looking window experiments (+0.024 local lift) to maintain strict rule compliance.

Milestone / Iteration	Key Technical Innovation	Local PR-AUC (Jul 1-15)	Kaggle Public LB
0. Baseline	Raw tabular features only, default LightGBM	0.1664	0.1664
1. Feature Pipeline	Expanding baselines, robust MAD z-scores, trailing windows	0.5090	0.5105
2. Objective Calibration	Removed <code>scale_pos_weight</code> (rank optimization)	0.5204	0.5395
3. Feature Pruning	Information-density pruning to Top-160 core features	0.5269	—
4. Dual-Horizon Trees	48% Full-Train (882r) + 52% 90-Day Specialist (433r)	0.5284	—
5. Tree-Neural Fusion	88% Dual-Horizon LGBM + 12% Tabular ResNet	0.5298	—
6. Temporal Propagation	Sliding LOO temporal propagation ($\pm 60\text{m}$, $w = 0.50$)	0.5359	0.56548